

1(a)(i)	width of red area after soaking or width of red area remaining ;	1												
1(a)(ii)	axes correctly labelled for independent <u>and</u> dependent variable ; line / smooth curve, showing inverse correlation ;	2												
1(b)(i)	<i>any seven from:</i> 1 use, same / same age, beetroot ; 2 minimum 5 cubes with different, size / dimensions ; 3 widths evenly spaced / if stated must be within range 0.5–5 cm ; 4 equipment for cutting / measurement of cubes ; 5 calculate / AW, the SA:V ratio ; 6 cubes are fully, covered / submerged, (by calcium hypochlorite) ; 7 remove after, same / stated, time (minutes) ; 8 method of maintaining the same temperature (once cubes are immersed) ; 9 method of drying cubes (after soaking) ; 10 cut in half and measure the width of the red tissue after soaking ; 11 repeat experiment a minimum of twice <u>and</u> calculate a mean ; 12 named hazard <u>and</u> risk <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>calcium hypochlorite</td><td>Irritant / harmful / toxic / corrosive</td><td>gloves / goggles PPE / tweezers</td></tr> <tr> <td>scalpel</td><td>cuts / injury, to <u>skin</u></td><td>cut away from fingers / cut on a tile</td></tr> <tr> <td>beetroot</td><td>allergen</td><td>gloves</td></tr> </tbody> </table>	hazard	risk	precaution	calcium hypochlorite	Irritant / harmful / toxic / corrosive	gloves / goggles PPE / tweezers	scalpel	cuts / injury, to <u>skin</u>	cut away from fingers / cut on a tile	beetroot	allergen	gloves	7
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Question	Answer	Mark
1(b)(ii)	1 $\frac{\text{Initial volume of (red) cube} - \text{volume final red cube}}{\text{Initial volume of (red) cube}} ;$ 2 $\times 100 ;$	2
1(c)(i)	17% (student 4 at 65°C) ; mean given of 89.3 at 25°C ;	2
1(c)(ii)	any two from: 1 all (cubes of) same, size / volume / dimension / number / SA:V ; 2 use, same / same age of, beetroot ; 3 sufficient water to cover the cubes completely ; 4 same volume of water in each beaker ; 5 same colorimeter ; 6 stir / AW, before taking sample for colorimeter reading ; 7 check / AW, temperature of water in beakers have equilibrated ; 8 zero colorimeter / calibrate, before each readings ; 9 use thermostatically controlled water-bath ;	2
1(c)(iii)	1 more (betalain) <u>released</u> as temperature increased or as temperature increases more betalain released ; max 1 from: 2 membrane, fluidity / permeability / AW, increases as temperature increases ; 3 membrane proteins denature, at high(er) temperatures ;	2